

Health Endpoint	Basis for Estimate *	Age Range		Unit Value	Unit Value Distribution	Parameters of Distribution	
		Min	Max			P1	P2
Worse; One or more symptoms; Shortness of Breath; Wheeze							

* All unit values pulled from a lognormal distribution from Model 1, Table III in Dickie and Ulery, 2002, are multiplied by 0.973811 to adjust for a difference in mean household income between the study participants and the general population. The unit values shown here have already been adjusted.

I.4.5 Upper Respiratory Symptoms (URS) in Children

In past benefit analyses, EPA based willingness to pay to avoid a day of URS on symptom-specific WTPs to avoid those symptoms identified as part of the URS complex of symptoms. Pope et al. (1991) defined a day of URS as consisting of one or more of the following symptoms: runny or stuffy nose; wet cough; and burning, aching, or red eyes. The three contingent valuation (CV) studies shown in Table I-8 have estimated WTP to avoid various morbidity symptoms that are either within the URS symptom complex defined by Pope et al., or are similar to those symptoms.

Table I-8. Median WTP Estimates and Derived Midrange Estimates (in 2015\$)

Symptom*	Dickie et al.	Tolley et al. (1986)	Loehman et al. (1979)	Mid-Range Estimate
Throat congestion	6.84	29.65	-	18.14
Head/sinus congestion	7.98	31.94	14.87	18.14
Coughing	2.29	25.11	9.03	12.70
Eye irritation	-	28.50	-	28.50
Headache	2.29	45.63	-	18.14
Shortness of breath	0.00	-	19.16	9.06
Pain upon deep inhalation (PDI)	8.01	-	-	8.01
Wheeze	4.57	-	-	4.57
Coughing up phlegm	4.99	-	-	4.99
Chest tightness	11.42	-	-	11.42

* All estimates are WTP to avoid one day of symptom. Midrange estimates were derived by IEC (1993).

** 10% trimmed mean.

The three individual symptoms that were identified as most closely matching those listed by Pope et al. for URS are cough, head/sinus congestion, and eye irritation, corresponding to “wet cough,” “runny or stuffy nose,” and “burning, aching or red eyes,” respectively. A day of URS could consist of any one of the seven possible “symptom complexes” consisting of at least one of these three symptoms. The original unit value for URS was based on the assumption that each of these seven URS complexes is equally likely. This unit value for URS, \$33.91, is just an average of the seven estimates of mean WTP for the different URS complexes. This unit value can be found in Table I-5. In

addition to the one unit value that EPA used in past benefit analyses, two other unit values can be found in BenMAP. Recent research by Dickie and Ulery (2002) suggests that parental WTP to avoid symptoms and illnesses in their children is about twice what it is to avoid those symptoms and illnesses in themselves. This second unit value of \$67.82 ($=2 \times \33.91) is derived from the unit value found in Appendix H, Section H.3.4.

Another unit value was derived by using Model 1, Table III in Dickie and Ulery (2002) (the same model used for acute bronchitis), assuming that a day of URS consists of 2 symptoms. As noted above, this model relates parental WTP to the number of symptom-days avoided and to whether it is the parent or the child at issue. The unit value derived from this model is \$264.

A WTP estimate elicited from parents concerning their WTP to avoid symptoms in their children may well include some calculation of lost earnings resulting from having to lose a day of work. Estimates from the Dickie and Ulery model therefore (appropriately) probably include not only their WTP to have their children avoid the pain and suffering associated with their illness, but also the opportunity cost of a parent having to stay home with a sick child. Unit values can be found in Table I-5.

I.4.6 Lower Respiratory Symptoms (LRS) in Children

The three unit values for LRS in children currently available in BenMAP follow the same pattern as those for URS in children. In past benefit analyses, EPA based willingness to pay to avoid a day of LRS on symptom-specific WTPs to avoid those symptoms identified as part of the LRS complex of symptoms. Schwartz et al. (1994) defined a day of LRS as consisting of at least two of the following symptoms: cough, chest tightness, coughing up phlegm, and wheeze. Of the symptoms for which WTP estimates are available (listed in Table H-8), those that most closely match the symptoms listed by Schwartz et al. are coughing, chest tightness, coughing up phlegm, and wheeze. A day of LRS, as defined by Schwartz et al., could consist of any one of 11 possible combinations of at least two of these four symptoms. In the absence of any further information, each of the 11 possible “symptom clusters” was considered equally likely. The unit value for LRS, \$21.43, is just an average of the eleven estimates of mean WTP for the different LRS symptom clusters. In addition to the original value used by the EPA in past benefits analyses, BenMAP contains a second unit value is twice the original unit value, or \$42.86. This value is based on the evidence from Dickie and Ulery (2002) that parents are willing to pay about twice as much to avoid symptoms and illness in their children as in themselves. The third unit value is based on Model 1, Table III in Dickie and Ulery, assuming that, as for URS, a day of LRS consists of 2 symptoms. As noted above, this model relates parental WTP to the number of symptom-days avoided and to whether it is the parent or the child at issue. The unit value derived from this model is \$264. These two additional unit values can be found in Table I-5.

I.4.7 Any of 19 Respiratory Symptoms

The presence of “any of 19 acute respiratory symptoms” is a somewhat subjective health effect used by Krupnick et al. (1990). Moreover, not all 19 symptoms are listed in the Krupnick et al. study. It is therefore not clear exactly what symptoms were included in the study. Even if all 19 symptoms were known, it is unlikely that WTP estimates could be obtained for all of the symptoms. Finally, even if all 19 symptoms were known and WTP estimates could be obtained for all 19 symptoms, the assumption of additivity of WTPs becomes tenuous with such a large number of symptoms. The likelihood that all 19 symptoms would occur simultaneously, moreover, is very small.

Acute respiratory symptoms must be either upper respiratory symptoms or lower respiratory symptoms. In the absence of further knowledge about which of the two types of symptoms is more likely to occur among the “any of 19 acute respiratory symptoms,” we assumed that they occur with equal probability. Because this health endpoint may also consist of combinations of symptoms, it was also assumed that there is some (smaller) probability that upper and lower respiratory symptoms occur together. To value avoidance of a day of “the presence of any of 19 acute respiratory symptoms” we therefore assumed that this health endpoint consists either of URS, or LRS, or both. We also assumed that it is as likely to be URS as LRS and that it is half as likely to be both together. That is, it was assumed that “the presence of any of 19 acute respiratory symptoms” is a day of URS with 40 percent probability, a day of LRS with 40 percent probability, and a day of both URS and LRS with 20 percent probability. Using the point estimates of WTP to avoid a day of URS and LRS derived above, the point estimate of WTP to avoid a day of “the presence of any of 19 acute respiratory symptoms” is:

$$(0.40)(\$33.91) + (0.40)(\$21.43) + (0.20)(\$33.91 + \$21.43) = \$33.20.$$

Because this health endpoint is only vaguely defined, and because of the lack of information on the relative frequencies of the different combinations of acute respiratory symptoms that might qualify as “any of 19 acute respiratory symptoms,” the unit dollar value derived for this health endpoint must be considered only a rough approximation. The value for any of 19 respiratory symptoms can be found in Table I-5.

Appendix J. Population & Other Data in U.S. Setup

This section describes the population, monitor, and demographic data in the United States setup. It consists of the following three subsections:

- **Population Data.** This describes how BenMAP forecasts population; the block-level and county-level data underlying the forecasts; and the PopGrid software application, which aggregates block-level population data to whatever grid definition might be needed.
- **Monitor Data.** The default United States setup has ozone, PM_{2.5}, PM₁₀, lead, NO₂, and SO₂ monitor data for the years 2000-2007. Data for CO are available at the BenMAP website: <http://www.epa.gov/air/benmap/>.
- **Demographic Datasets.** This subsection describes the various datasets in the U.S. setup related to demography: household size, poverty rates, and educational attainment.

J.1 Population Data in U.S. Setup

The U.S. setup in BenMAP calculates health impacts for any desired grid definition, so long as you have a shapefile for that grid definition and population data for that grid definition. In this description, we use the term “population grid cell” to refer to a cell (e.g., county) within a grid definition. The foundation for calculating the population level in the population grid-cells is 2020 Census block data. A separate application called “PopGrid,” described below, combines the Census block data with any user-specified set of population grid-cells, so long as they are defined by a GIS shape file. Unfortunately, PopGrid relies on extremely large census files that are too large to include with BenMAP -- hence the need for the separate application. The PopGrid program is available on U.S. EPA’s BenMAP GitHub organization here: <https://github.com/BenMAPCE/PopGrid2>.

Within any given population grid-cell, BenMAP has 304 unique race-ethnicity-gender-age groups: 19 age groups by 2 ethnic groups by gender by 4 racial groups (19*2*2*4=304). Table J-1 presents the 304 population variables available in BenMAP. As discussed below, these variables are available for use in developing age estimates in whatever grouping you require.

Table J-1. Demographic Groups and Variables Available in BenMAP

Racial Group	Ethnicity	Age	Gender
White, African American, Asian, American Indian	Hispanic, Non-Hispanic	<1, 1-4, 5-9, 10-14, 15-19, 20-24, 25-29, 30-34, 35-39, 40-44, 45-49, 50-54, 55-59, 60-64, 65-69, 70-74, 75-79, 80-84, 85+	Male, Female

In this section on population data in the U.S. setup, we describe:

- **Forecasting Population.** This describes how BenMAP forecasts population.